Session 10: CNNs, from ImageNet to Biology

KB8029 — Bioinformatics & Deep Learning

Arne Elofsson

SciLifeLab

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Log in to the lecture quiz

Log in to the quiz

Scan the code, or go to quiz.bioinfo.se/live

Log in with your university account

Answering counts as being at the lecture: up to 3 bonus points on the exam

Right or wrong is never graded, only whether you answered. Results are shown without names.

quiz.bioinfo.se/live

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Warm-up: what do you know already?

3 quick questions

On the topics of today, one at a time, about 30 seconds each

Answer on your own, and guess if you do not know

It shows me where to start; right or wrong is never graded

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A 3 × 3 kernel slides over a 6 × 6 image, without padding, one step at a time. How large is the output?

A 6 × 6

B 4 × 4

C 3 × 3

D 2 × 2

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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A filter, 8 bases wide, slides along one-hot encoded DNA. What is it most like?

A A BLAST alignment

B A phylogenetic tree

C A position weight matrix

D A restriction enzyme

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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What does transfer learning mean?

A Moving the model from the CPU to a GPU

B Starting from a network trained on another, larger data set

C Copying labels from similar proteins

D Training two networks and averaging them

Answer now
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quiz.bioinfo.se/live

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What the room knows already

Where we start

Which topics most of you know, and which we take slowly today.

Show the results ▸ quiz.bioinfo.se/live/report

Your questions before today

What you asked

The questions you sent in on Canvas before the lecture

The most common ones now; the others in the block where they belong

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In up to three words: what do you want to learn today?

Why convolution

weight sharing, equivariance

Sequences

filters as motifs, pooling

Images

transfer, augmentation, contact maps

Answer now

Scan the code, or go to
quiz.bioinfo.se/live

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What you want to learn today

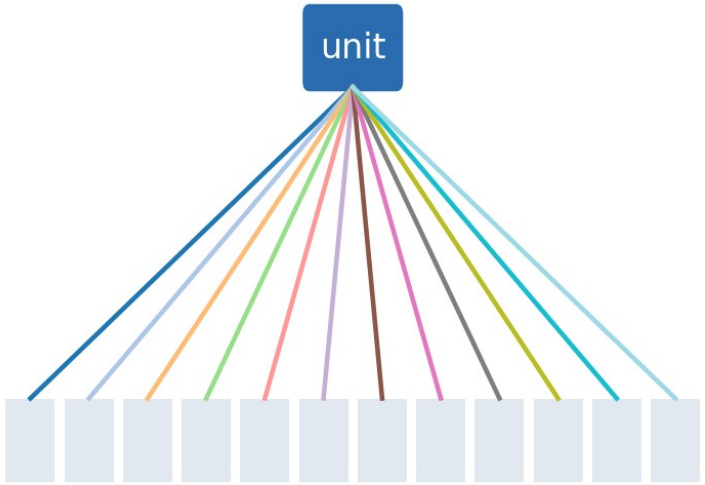
What the room wants to learn

Show the results ▸ quiz.bioinfo.se/live/report

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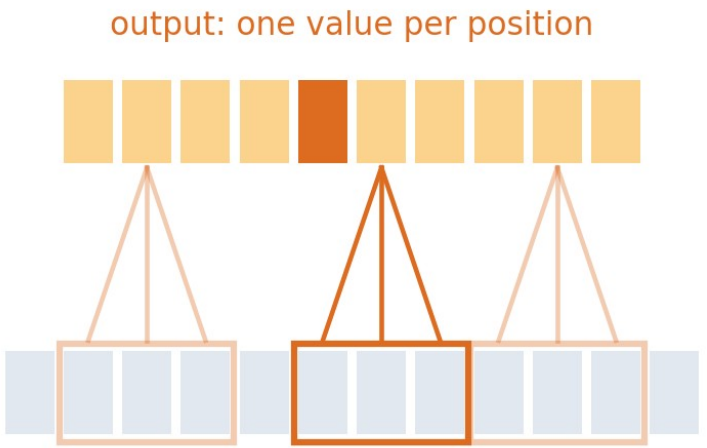
One small detector, reused everywhere

Dense: a weight for every position



the same pattern must be learned again at every position

Convolution: one small detector, reused



3 shared weights, whatever the length

A dense layer learns a separate weight for every position. A convolutional layer assumes that nearby inputs belong together and reuses the same small detector at every position.

See course.bioinfo.se/session10.html.

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Three blocks: about 30 min talk and 15 min discussion



Block 1: where CNNs and GPUs came from, then why convolution works. Block 2: CNNs on DNA and protein sequences. Block 3: CNNs on images, from cancer histology and contact maps to an outlook: the Human Protein Atlas and Alpha Cell.

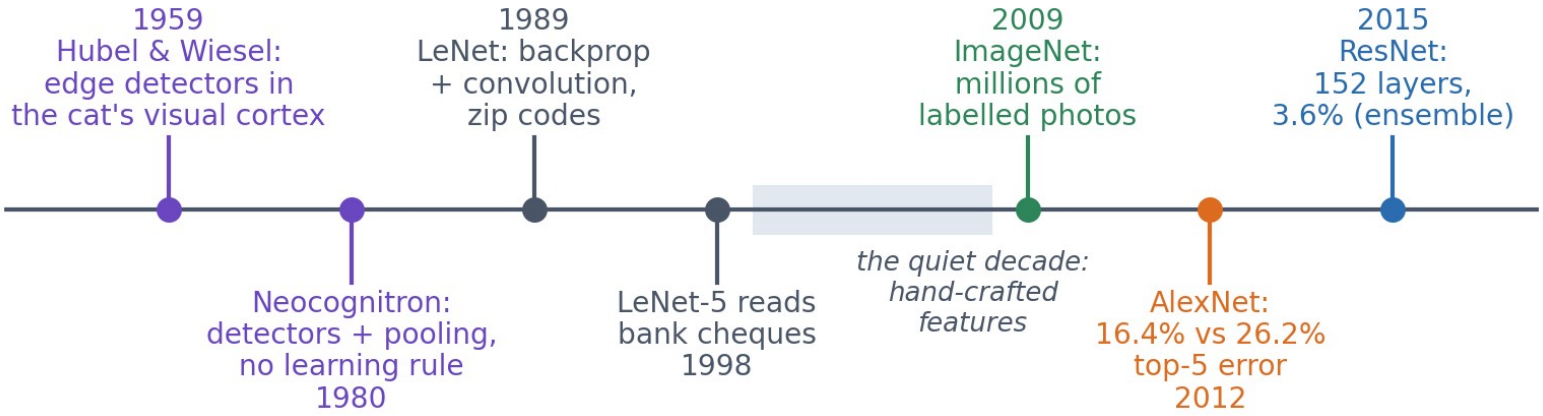
Times are approximate: a 10 min start (warm-up quiz and your questions), two 15 min breaks and a 5 min close. Each block ends with a quiz question: guess, run the notebook and work out why it behaves as it does, discuss, answer again.

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BLOCK 1 · ABOUT 30 MIN

Where CNNs came from, and why convolution works

From cat neurons to AlexNet: a 60-year-old idea

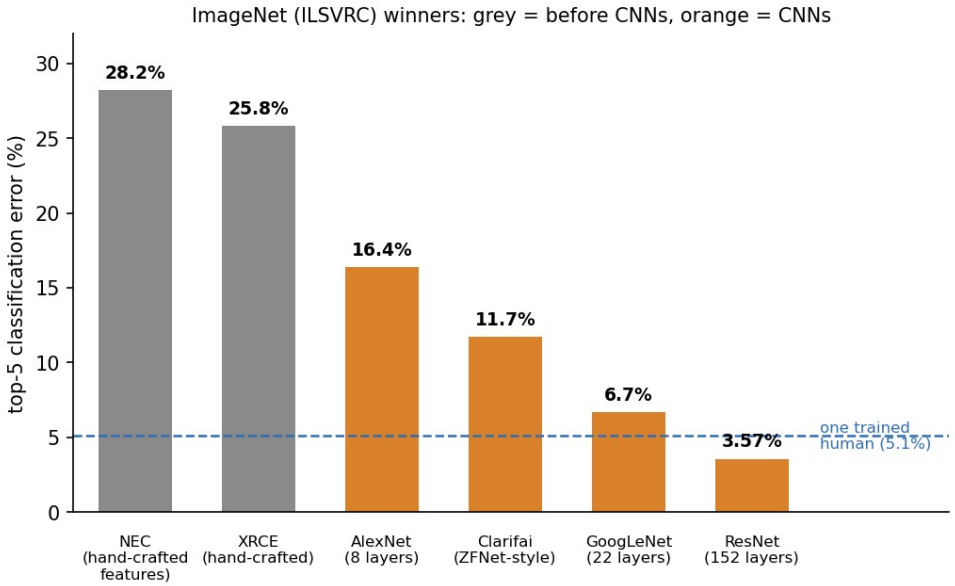


The two operations of a CNN, a local detector and pooling, were seen in the visual cortex in 1959 and built into a network in 1980. Backpropagation made them trainable in 1989. Then CNNs waited two decades for data and compute.

Hubel & Wiesel 1962; Fukushima 1980; LeCun et al. 1989, 1998; Deng et al. 2009; Krizhevsky et al. 2012; He et al. 2016. Not to scale in time.

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2012: AlexNet wins ImageNet by a mile



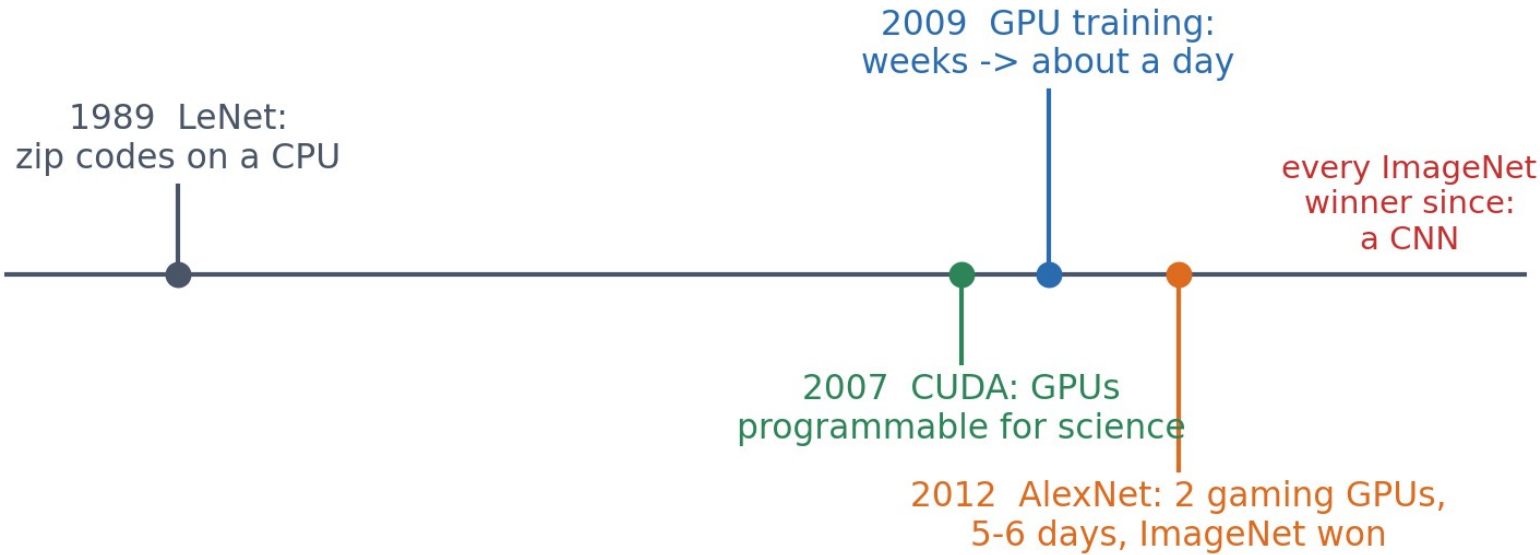
2012: 16.4% with the provided training data (15.3% with extra data); runner-up 26.2%.

- Krizhevsky, Sutskever & Hinton, Toronto
- 60 million weights, 2 gaming GPUs, 5-6 days
- **Top-5 error 16.4% vs 26.2%**
- Old parts: convolution, ReLU, dropout, augmentation
- **New: the scale of data + GPUs**

Russakovsky et al. 2015; Krizhevsky et al. 2012; He et al. 2016.

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The GPU revolution: from weeks to days

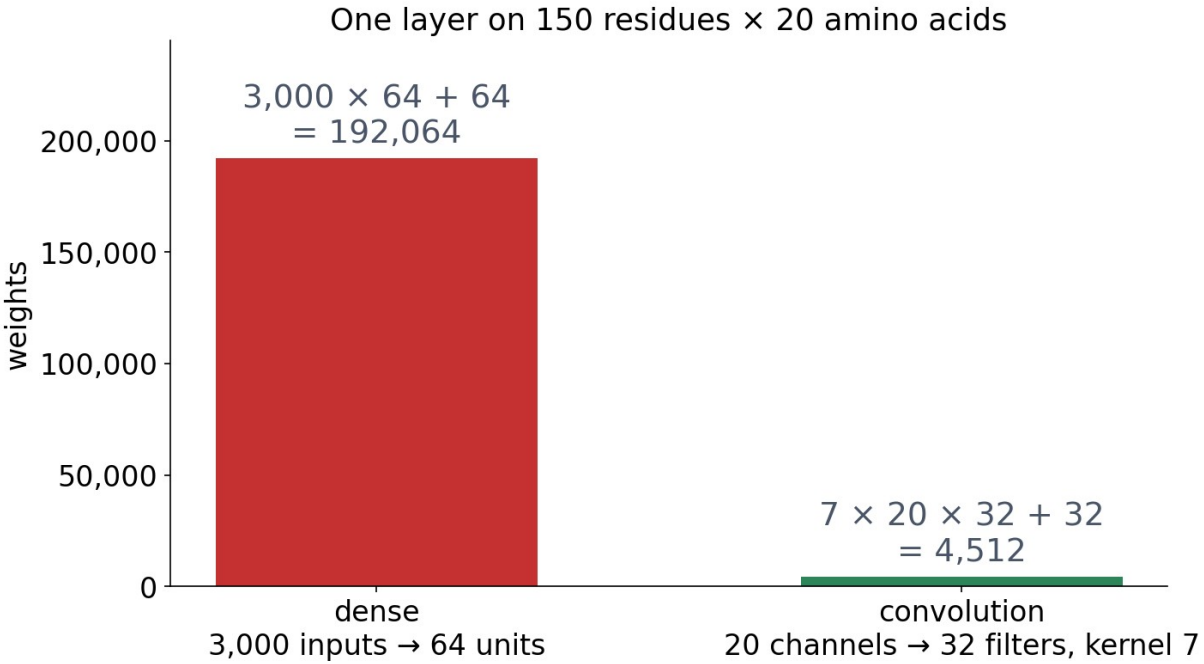


Training a network is mostly huge matrix products, and graphics cards have thousands of simple cores built for exactly that. AlexNet was trained on two gaming GPUs in five to six days; without GPUs there would be no deep CNNs.

Raina, Madhavan & Ng (2009), ICML; Krizhevsky, Sutskever & Hinton (2012), NeurIPS.

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The problem with dense layers



- **40× fewer weights**
- Same count for any length
- A 224 × 224 colour image: 150,000 inputs per unit

See course.bioinfo.se/session10.html, 'A small detector, reused everywhere'.

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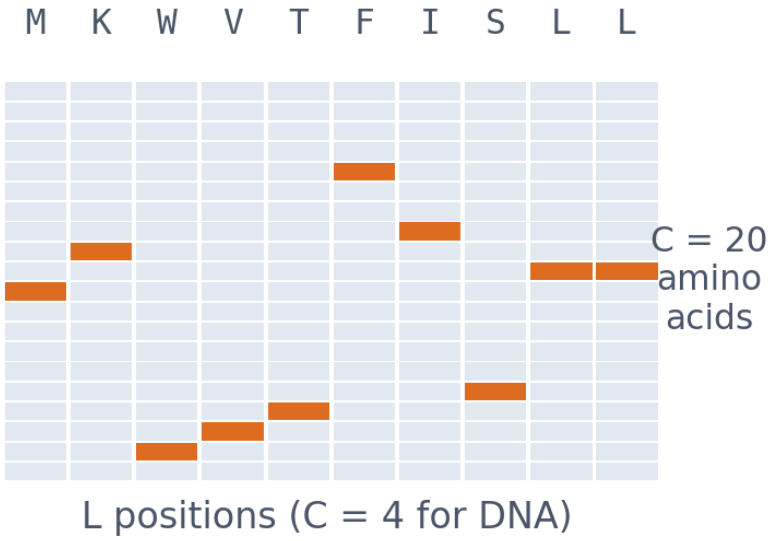
Shapes: images and sequences

image: B × C × H × W



C = 3 colour channels

sequence: B × C × L

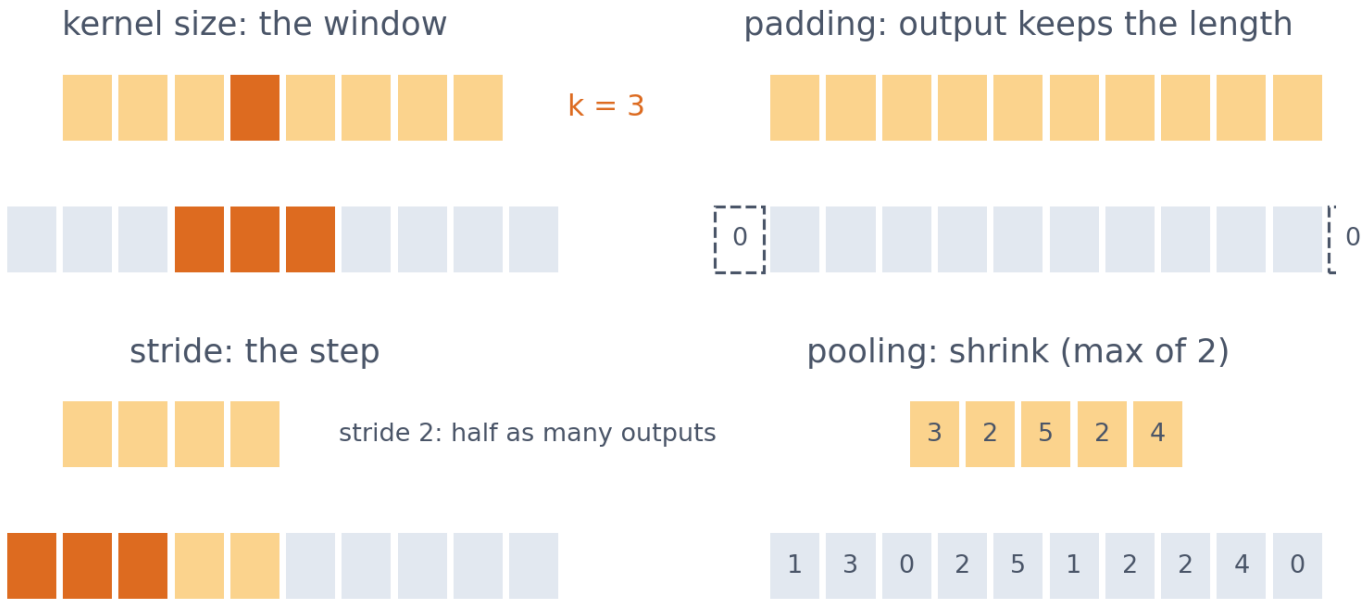


A batch of images is B × C × H × W; a batch of sequences is B × C × L. A size-7 kernel on protein therefore has 20 × 7 weights, one per amino acid per position.

See course.bioinfo.se/session10.html, 'Shapes'.

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Four settings of a convolutional layer



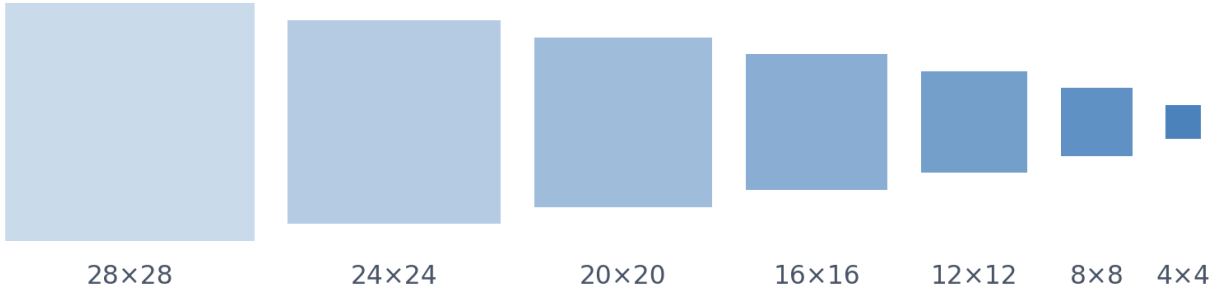
Kernel size: the width of the window. Padding: zeros at the edges, so the output keeps the input's length. Stride: how far the kernel moves. Pooling: the maximum or mean of small regions.

See course.bioinfo.se/session10.html, 'Shapes'.

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Without padding, every layer shrinks the image

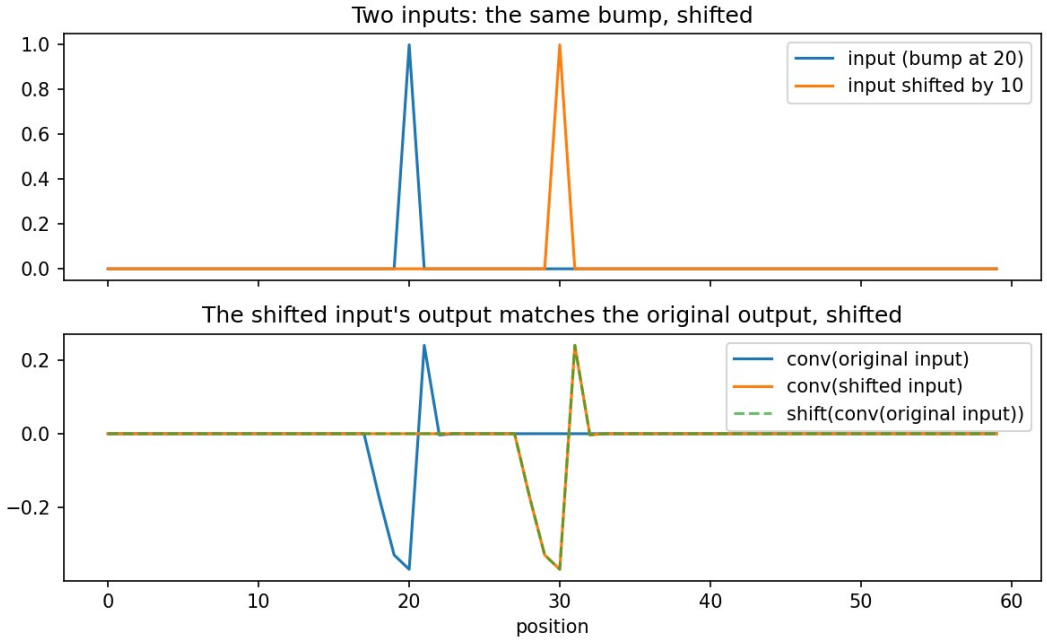
Six 5×5 layers without padding: 28 → 4; a seventh no longer fits



with padding = 2 ('same'): still 28×28 after 20 layers

- Kernel 5: lose 4 pixels
- **Padding keeps the size**
- Then depth is free

Shift the input, and the output shifts

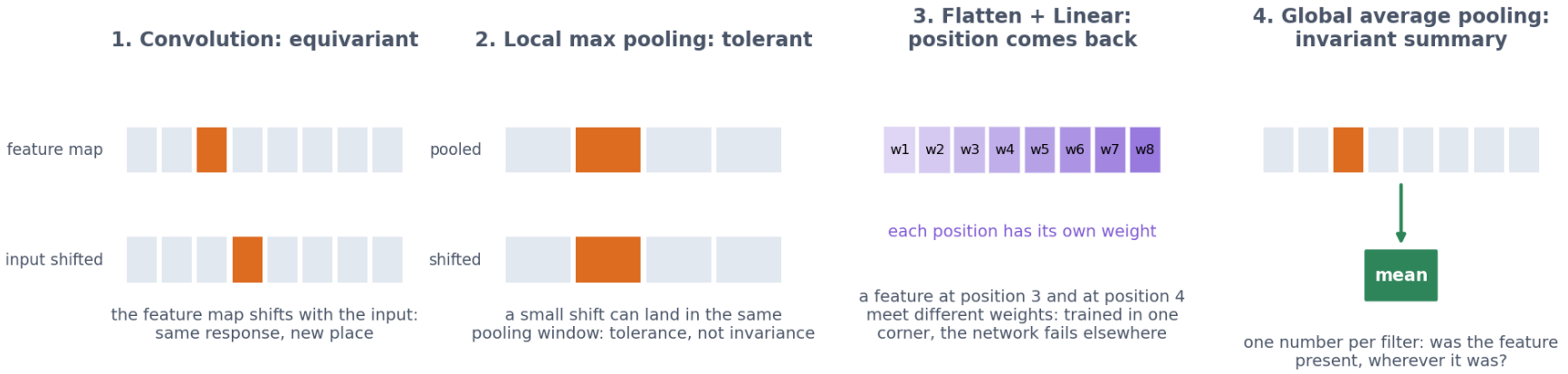


- **Comes from weight sharing**
- Holds even untrained
- **Equivariance ≠ invariance**

Course notebook session10-cnns-in-code.ipynb, Part 1.

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Where does position information go?

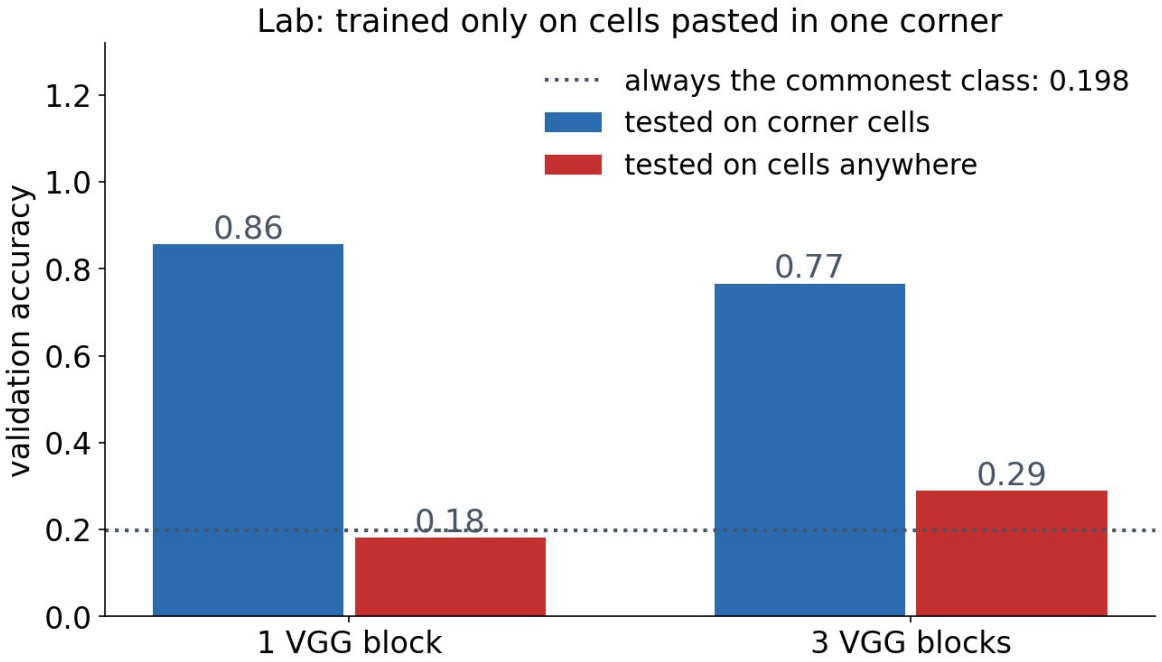


Convolution: the feature map shifts. Local pooling: tolerance to small shifts. Flatten + Linear: every position gets its own weight again. Global average pooling: one number per filter, wherever the feature was.

See course.bioinfo.se/session10.html, 'Equivariance is not invariance'.

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Flatten brings position back

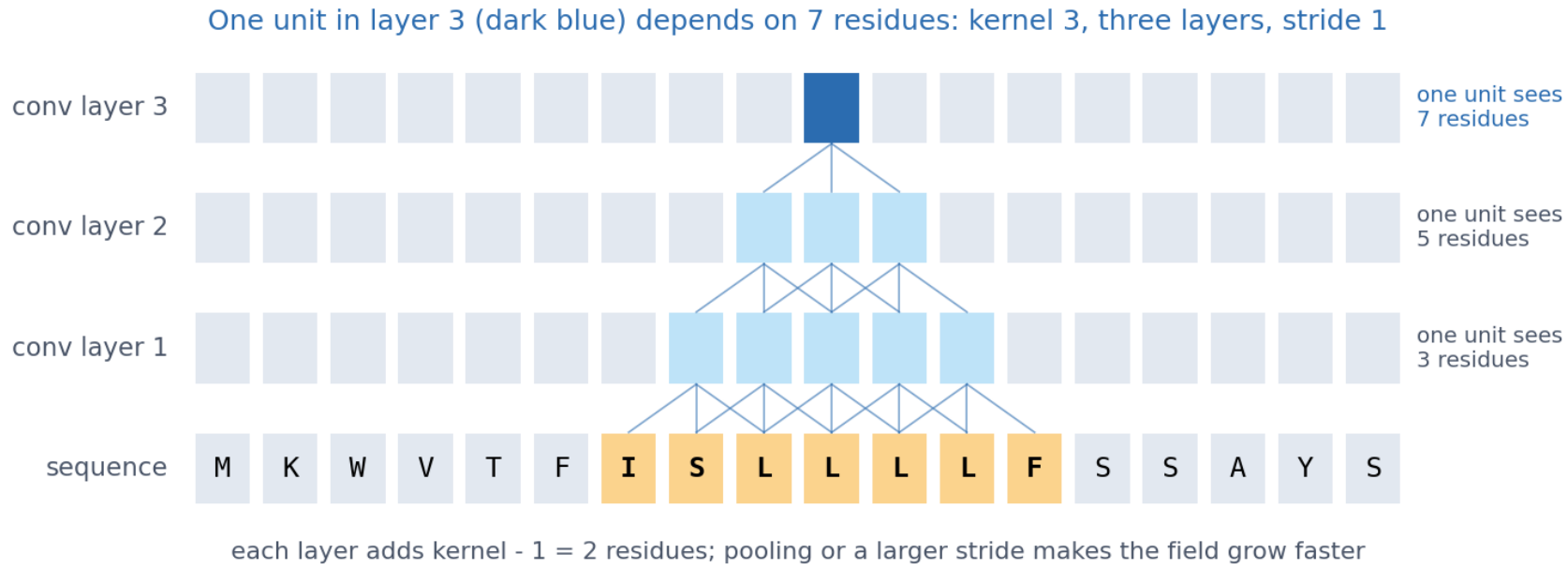


- Trained on corner cells only
- **Anywhere else: near chance**
- Fix: global pooling, or train on all positions

Session 10 lab, executed solution.

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The receptive field grows with depth



Kernel 3, stride 1: a unit in layer 1 sees 3 residues, in layer 2 five, in layer 3 seven. Each layer adds k − 1; pooling and stride make it grow faster.

See course.bioinfo.se/session10.html, 'Stacking convolutions: the receptive field'.

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How many weights does the convolutional layer have?

nn.Conv2d(3, 64, kernel_size=3, bias=False), on a 224 × 224 and on a 448 × 448 RGB image.

```
layer = nn.Conv2d(3, 64, kernel_size=3,
                  bias=False)
layer(image_224)   # 224 × 224, RGB
layer(image_448)   # 448 × 448, RGB
layer.weight.numel()  # = ?
```

A 1,728 for both images

B 1,728, then 6,912: four times the pixels

C 9,633,792, then 38,535,168

D 576 for both images

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live

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How many weights does the convolutional layer have?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

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How many weights does the convolutional layer have?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour



The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session10-discussion-1.ipynb

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How many weights does the convolutional layer have?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

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How many weights does the convolutional layer have?

Results, and who explains

Both rounds side by side.

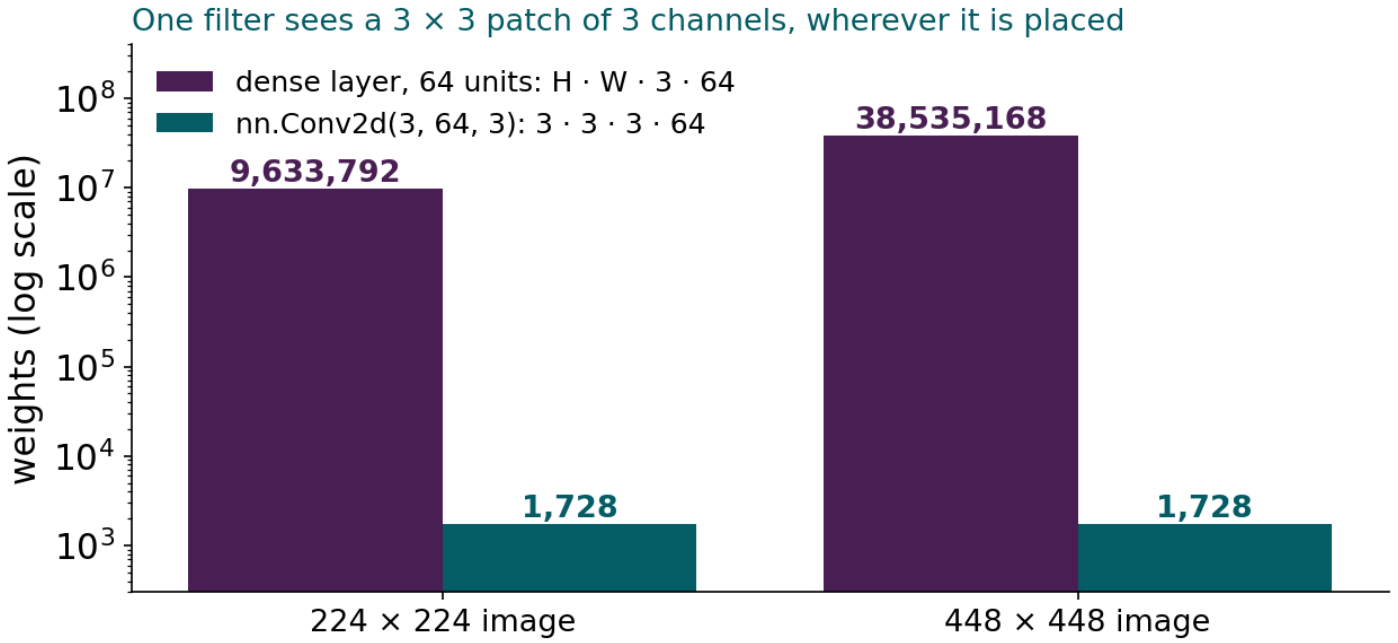
Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 1 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session10-discussion-1.ipynb

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The answer: 1,728 for both images



One filter looks at a 3 × 3 patch of the 3 colour channels, wherever it is placed: 27 weights, and 64 filters make 1,728. The image size never enters. A dense layer with 64 units needs one weight per pixel, channel and unit: 9.6 million, and four times as many when each side doubles.

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BLOCK 1 DONE

Break

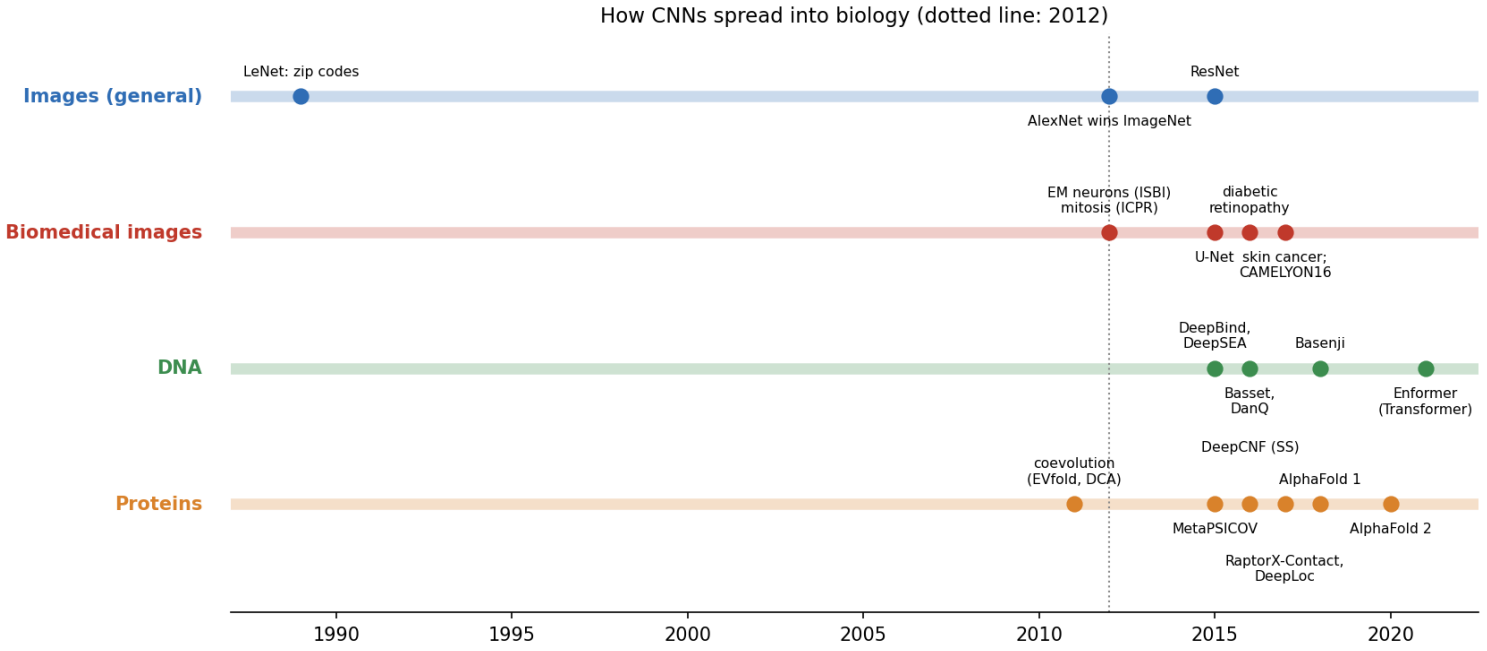
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BLOCK 2 • ABOUT 30 MIN

CNNs on biological sequences

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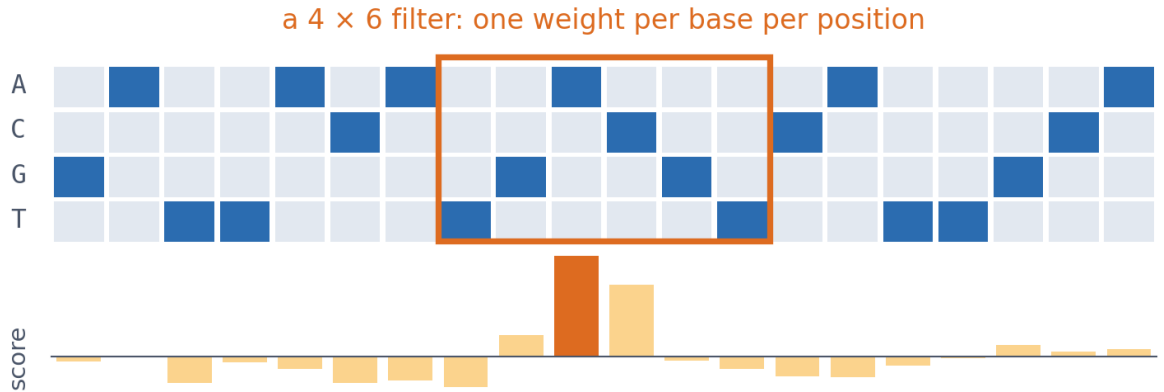
How CNNs spread into biology



DeepBind (2015): binding specificities visualised as PWMs. DeepSEA (2015): 1,000 bp of sequence, 919 chromatin features, and scores for non-coding variants. Later networks rediscovered known transcription-factor motifs.

Alipanahi et al. 2015; Zhou & Troyanskaya 2015.

A filter on DNA is a learned PWM



slide it along the sequence = scoring with a PWM (Session 5), but learned (schematic)

- **4 × k weights**
- Sliding it = scoring with a PWM
- Learned, not counted

See course.bioinfo.se/session10.html, 'Filters become motifs'.

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DeepSEA: 1,000 bases in, 919 answers out

- Reads 1,000 bp of sequence
- Predicts 919 chromatin features at once
- **A non-coding variant: predict the reference, predict the mutated sequence, compare**
- Longer range needed larger receptive fields, and eventually Transformers (Session 11)

Zhou & Troyanskaya 2015, Nat Methods 12:931. See course.bioinfo.se/session10.html.

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Our localisation data, as a 1D CNN

The localisation network of Sessions 8-9, as a 1D CNN

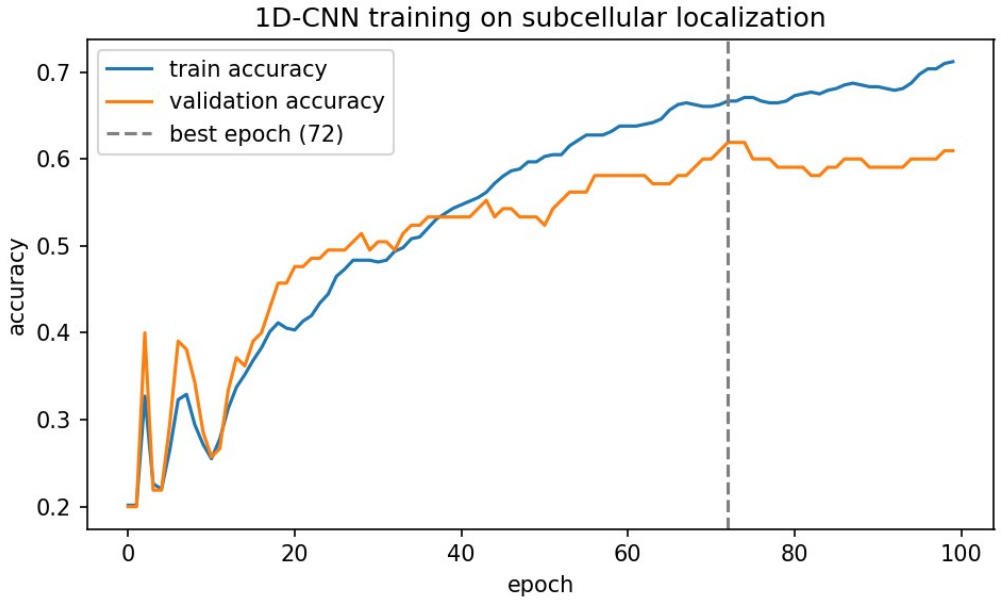


19,237 weights (flat network: 192,389)

- Same proteins as Sessions 8-9
- Same homology-aware split
- **A tenth of the weights**

Course notebook, Part 2.

Training with early stopping

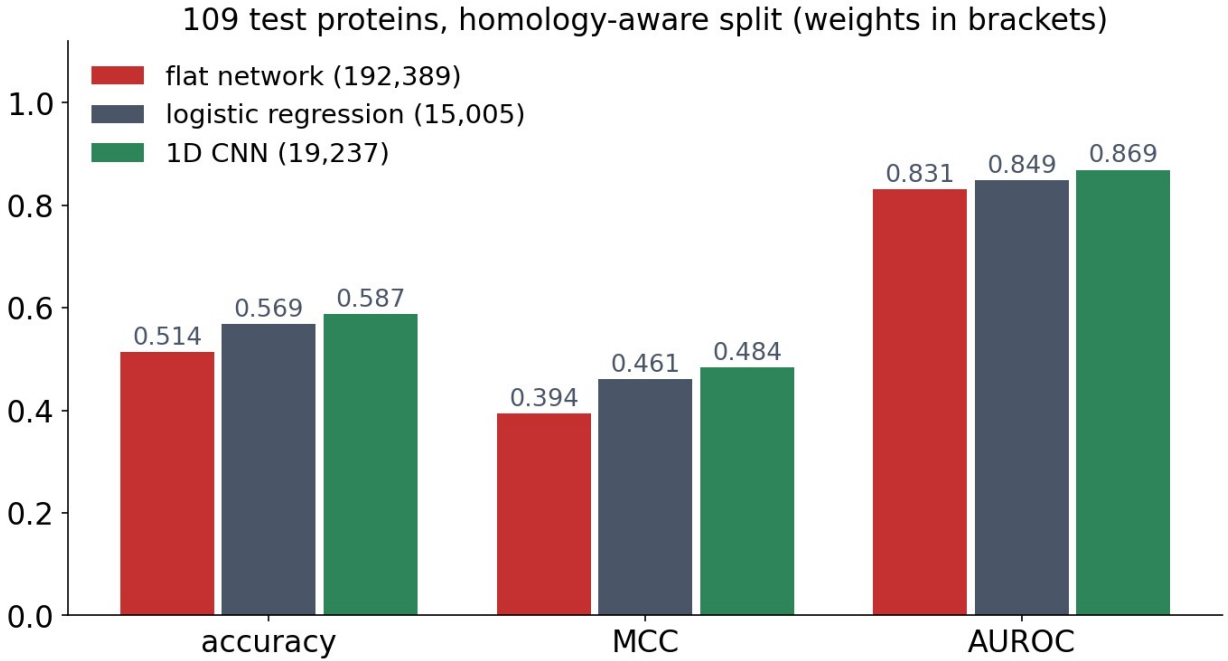


- Best validation: 0.619 at epoch 72
- **No longer memorises**
- Training accuracy then 0.667

Course notebook, Part 2.

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The CNN beats the flat network on every metric

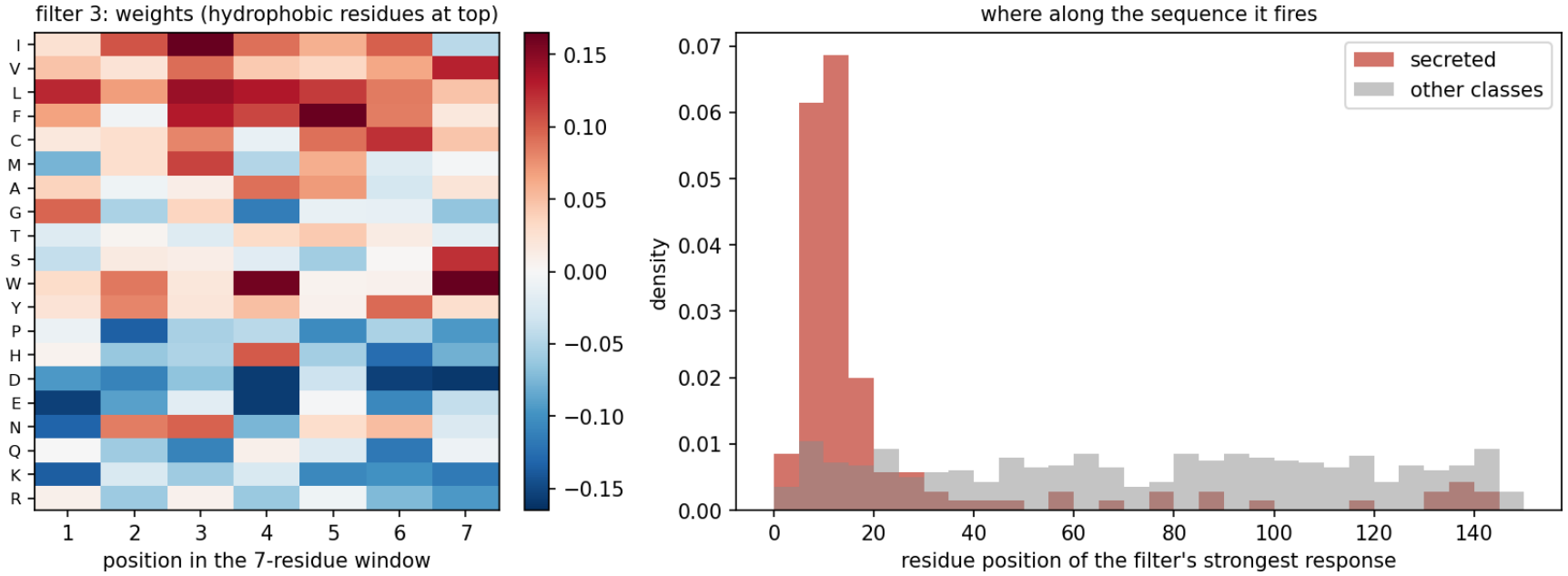


- **Built-in assumption, not 192,389 free weights**
- Margin over logistic regression: noise

Course notebook, Part 2; test set used once.

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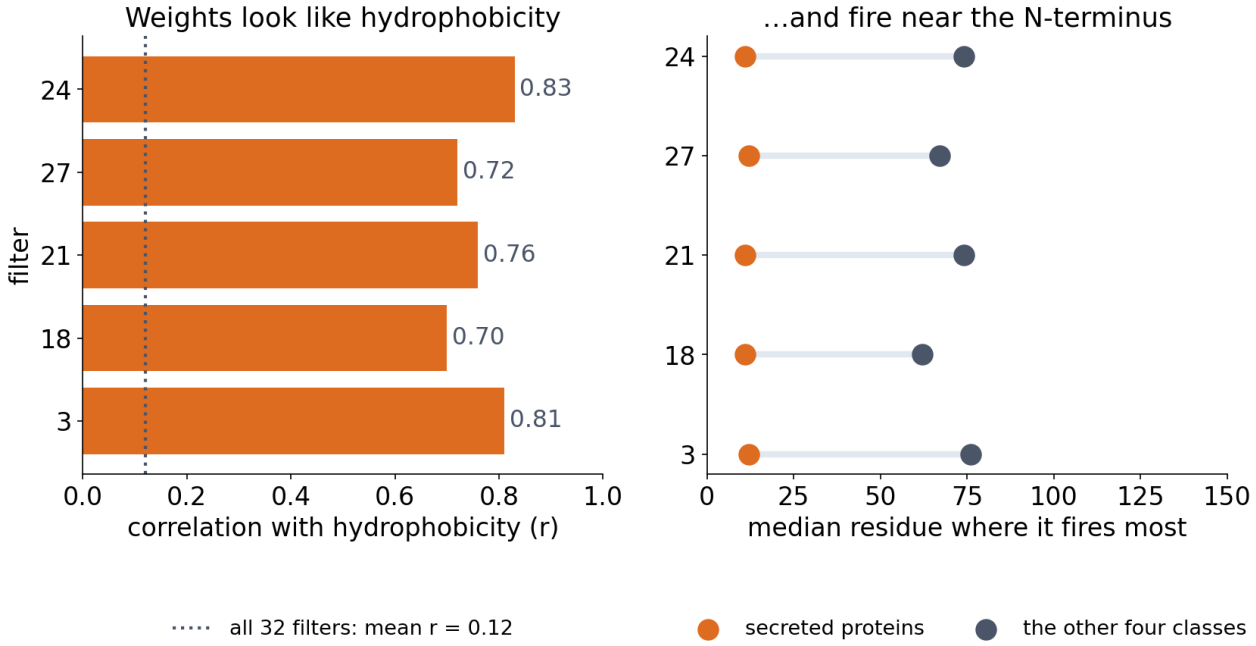
Inside the first layer: a learned signal peptide



Left: the weights of the filter most selective for secreted proteins, hydrophobic amino acids at the top. Right: where along the sequence it fires most strongly, in secreted proteins and in the other four classes.

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Nobody told it about signal peptides



The five filters most selective for secreted proteins: weights correlated with Kyte-Doolittle hydrophobicity (r = 0.70-0.83), firing around residue 11-12 in secreted proteins and 62-76 in the others.

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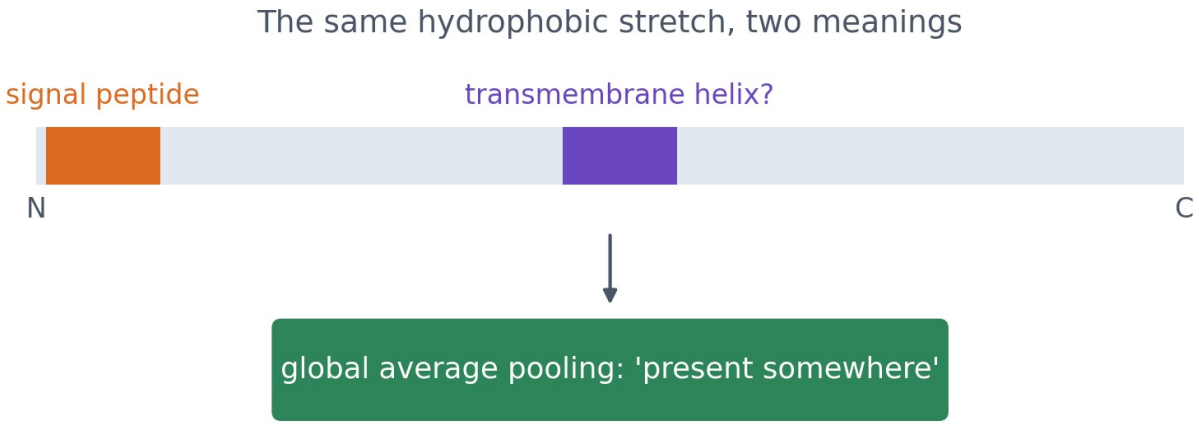
Its second layer sees about a signal peptide

- Layer 1: a 7-residue window
- Then 2× pooling, then another 7-wide kernel
- **Each layer-2 unit sees 20 residues**
- About the length of a signal peptide
- Discussion problem 2: how wide must the window be?

See course.bioinfo.se/session10.html, 'Proteins: our own CNN finds the signal peptide'.

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What has global pooling thrown away?



- **Detected the same everywhere**
- **Not the same meaning everywhere**
- How would you give position back?

See course.bioinfo.se/session10.html, 'But position matters in proteins'. Schematic.

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DeepLoc: convolutions first, order afterwards

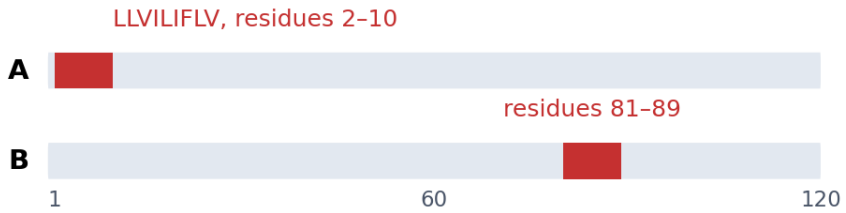
- Convolutional layers find the local patterns
- A bidirectional LSTM with attention then reads them in order
- **The order-aware half restores the position we pooled away**
- Session 11 builds that half

Almagro Armenteros et al. 2017, Bioinformatics 33:3387.

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What does the filter report after global pooling?

One filter, 7 residues wide, for L, I, V and F. The same hydrophobic stretch: in A at residues 2–10, in B at 81–89.



one filter, 7 residues wide: weight 1 for L, I, V, F, else 0

then global max pooling over the whole protein

what does the filter report for A and for B?

A The same value for both

B Higher for A: the stretch is at the N-terminus

C Higher for B: more sequence before it

D Zero for both: pooling removes the signal

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live

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What does the filter report after global pooling?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

SciLifeLab

What does the filter report after global pooling?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session10-discussion-2.ipynb

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What does the filter report after global pooling?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

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What does the filter report after global pooling?

Results, and who explains

Both rounds side by side.

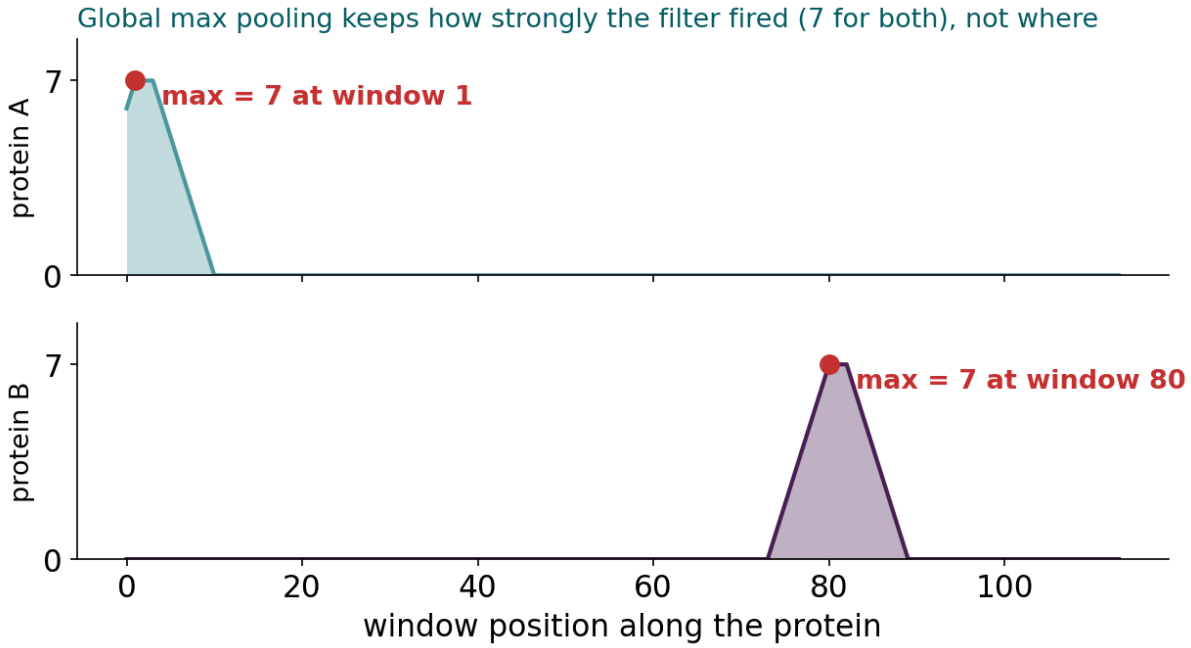
Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 2 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session10-discussion-2.ipynb

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The answer: the same value; pooling forgets where



The filter's response moves with the stretch, and its maximum is 7 in both proteins. Global pooling keeps how strongly a filter fired, not where, although in a protein the position matters: a hydrophobic stretch at the N-terminus suggests a signal peptide, in the middle a transmembrane helix.

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BLOCK 2 DONE

Break

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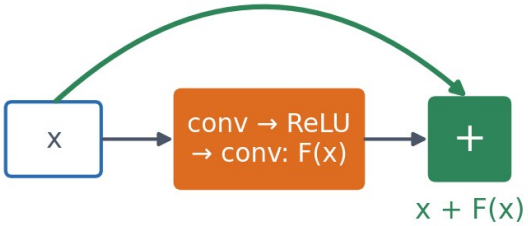
BLOCK 3 · ABOUT 30 MIN

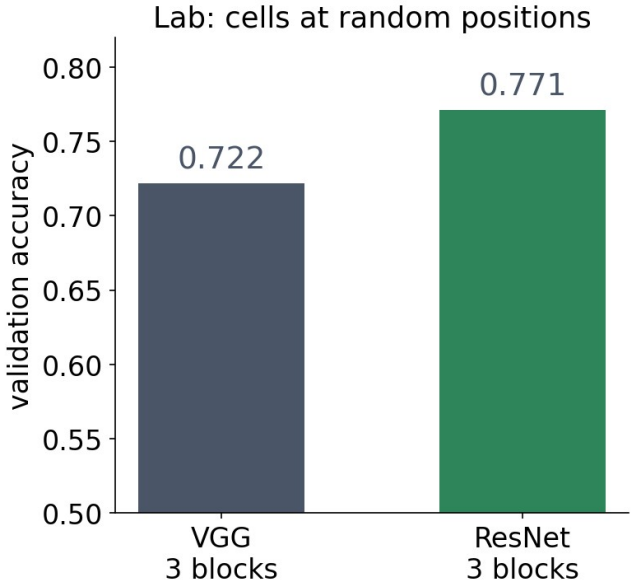
CNNs on images: from the clinic to the whole cell

ResNet: a short path for the gradient

A residual block (He et al. 2016)

shortcut: the gradient's short path



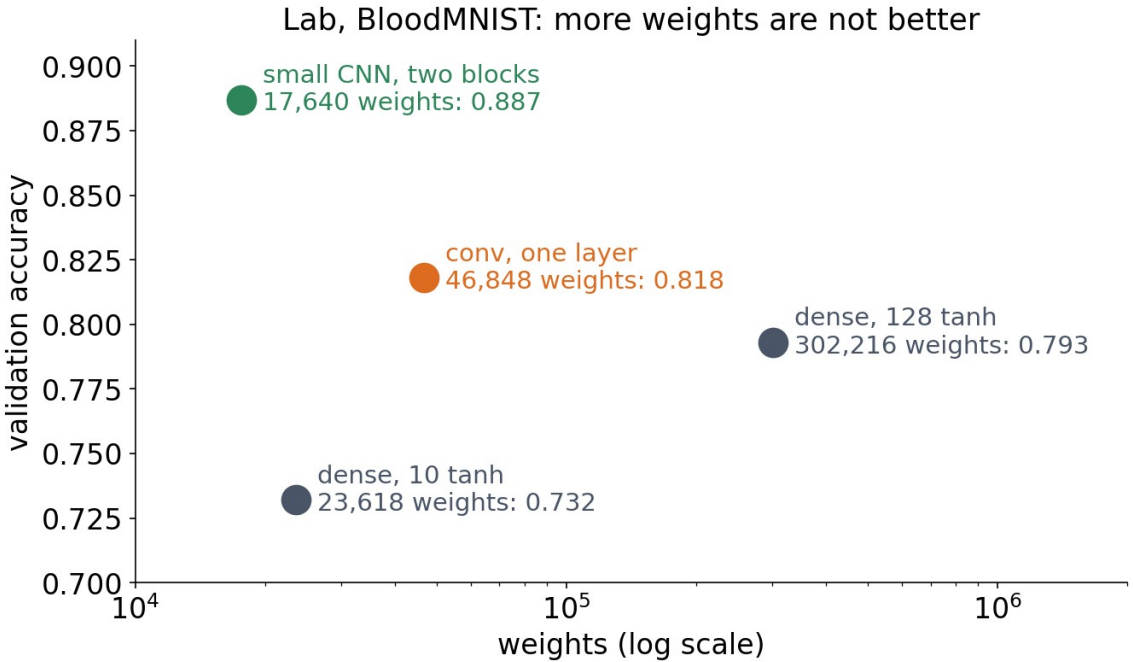


- **Each block returns x + F(x)**
- 100+ layers become trainable
- Deep plain networks were not overfitting

He et al. (CVPR 2016); Session 10 lab, optional extra.

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More weights are not better

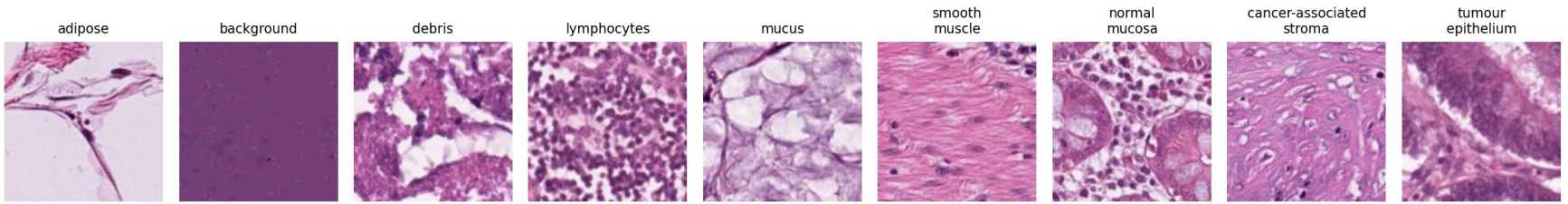


- **Smallest model, best result**
- Locality and sharing are built in
- Session 9's lesson, from the other side

Session 10 lab, executed solution (BloodMNIST).

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Colorectal cancer histology: nine tissue types

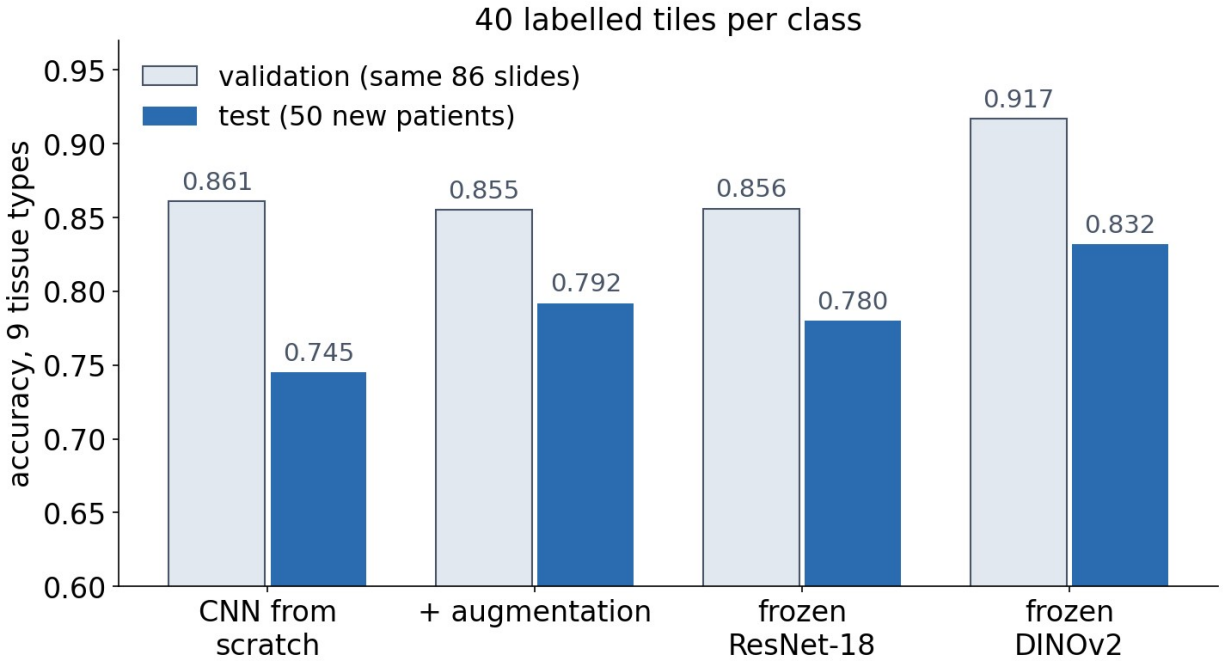


H&E-stained tiles (Kather et al. 2019). We pretend to have only 40 labelled tiles per class; validation tiles come from the same 86 slides, test tiles from 50 other patients.

NCT-CRC-HE-100K / CRC-VAL-HE-7K (Kather, Halama & Marx 2018, CC BY 4.0).

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Transfer learning beats training from scratch

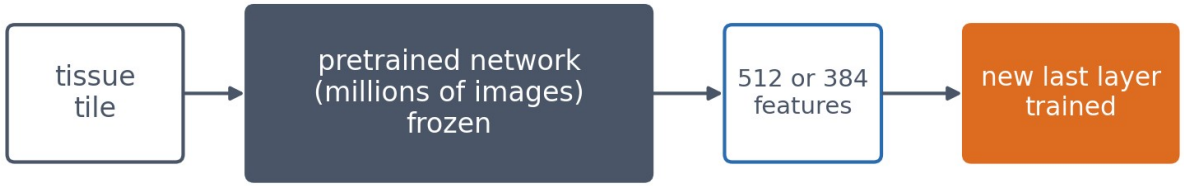


- **DINOv2 features: best**
- The seed alone moves it 6 points
- **Validation > test: split by patient**

Course notebook, Part 1. Test: 900 tiles from 50 patients.

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Borrow features learned on millions of images



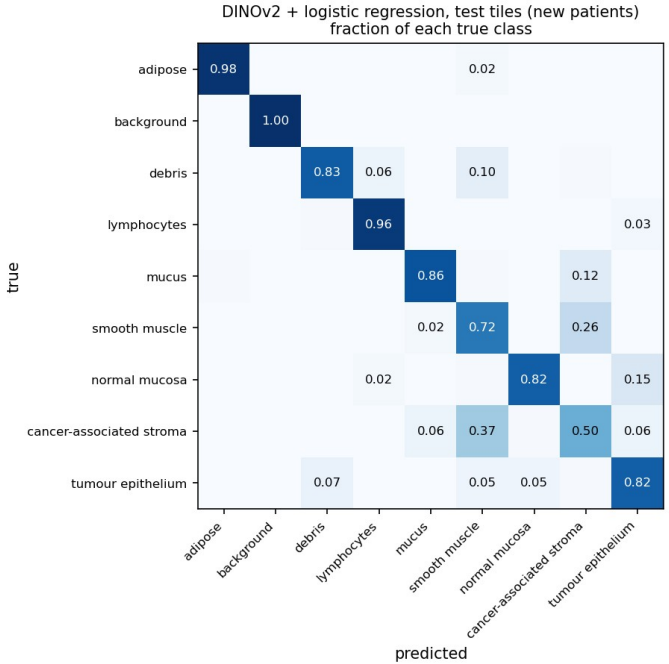
ResNet-18: ImageNet labels · DINOv2: 142 million images, no labels

9 tissue types

- Keep the network frozen
- **Train only a new last layer**
- A foundation model (Session 11)

Course notebook, Part 1; Oquab et al. 2023 (DINOv2).

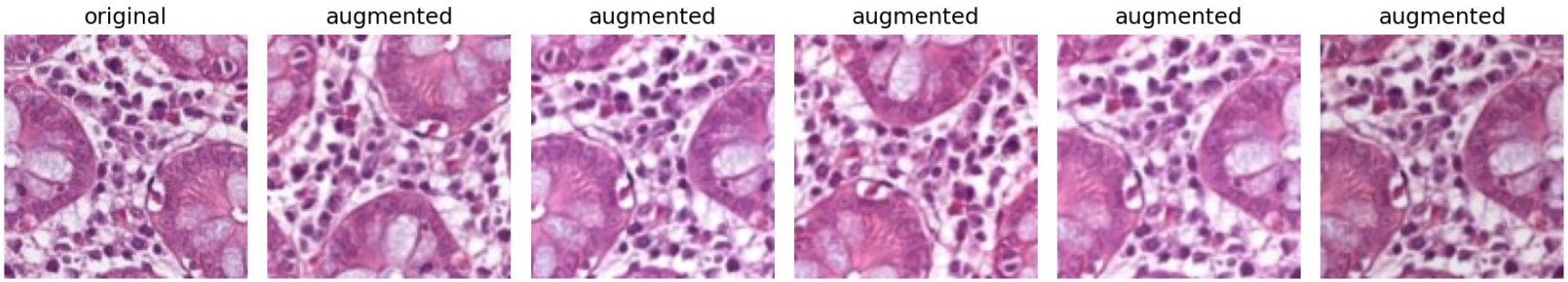
Which tissues does it still confuse?



- Stroma: 0.50 correct
- **Mistaken for smooth muscle**
- Normal mucosa ↔ tumour epithelium

Course notebook, Part 1: DINOv2 + logistic regression, test tiles.

Augment only what preserves the label

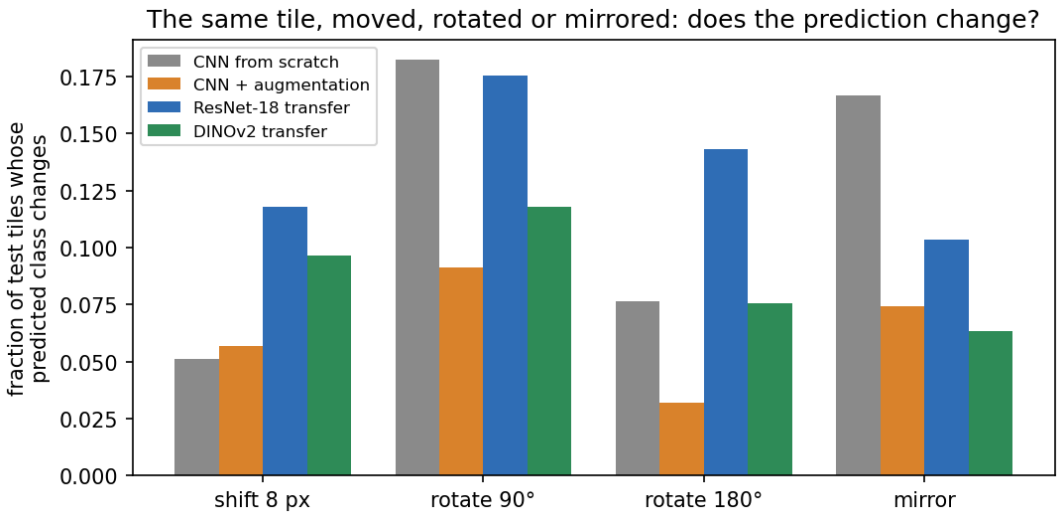


One training tile and five augmented copies. A tissue section has no 'up', so rotations and mirror images are valid, and small colour changes imitate staining differences. A 6 rotated by 180° is a 9: wrong for digits.

See course.bioinfo.se/session10.html, 'Which transformations should not matter?'.

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Rotation is not built in: it must be learned

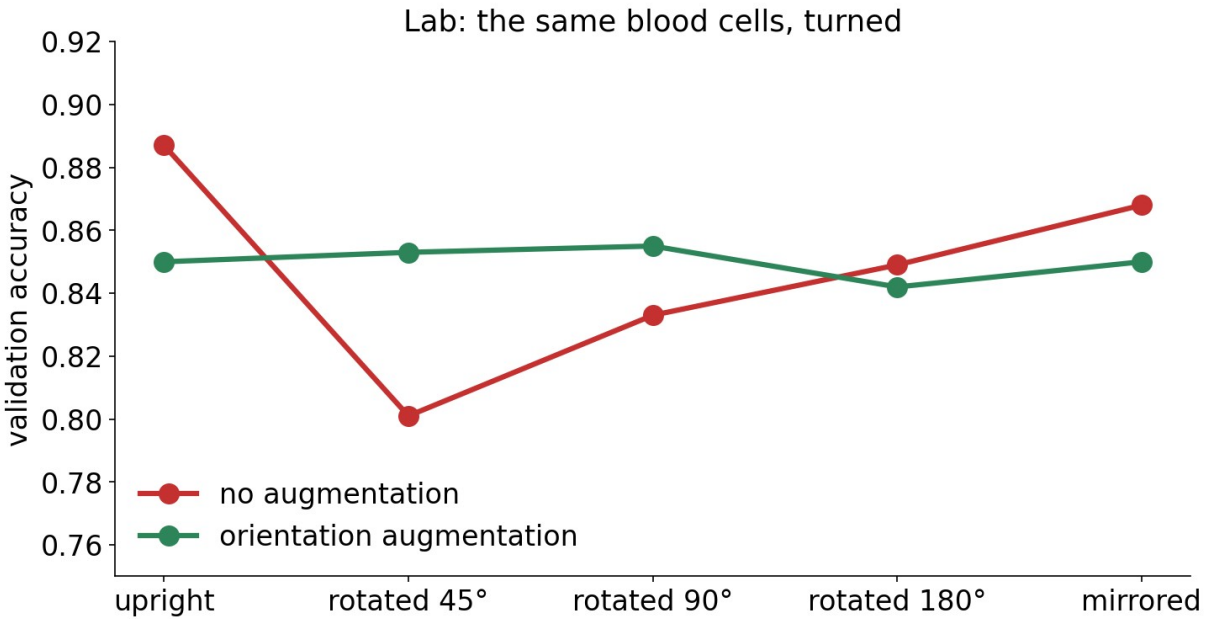


- Shift 8 px: 5% change
- **Rotate 90°: 18% change**
- **Rotate 90°, augmented: 9%**

Course notebook, Part 1; Veeling et al. 2018.

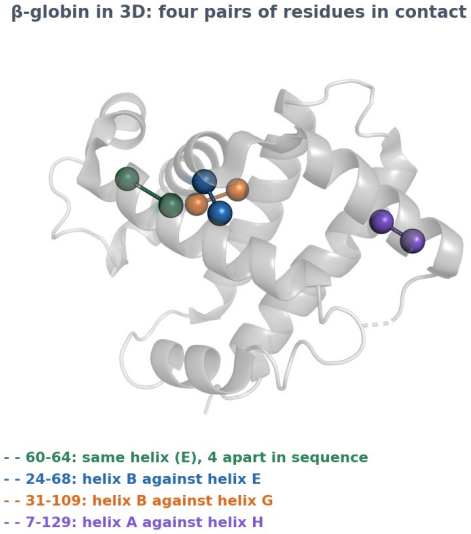
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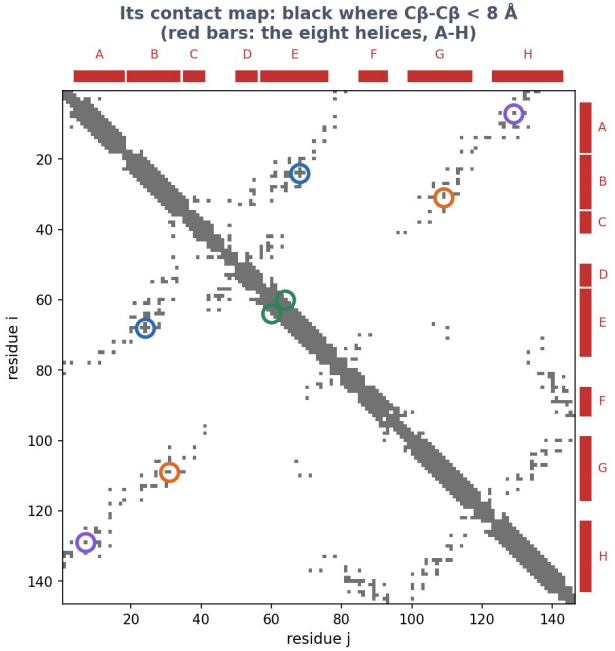
Turned blood cells: augmentation evens it out



- Without: worst at 45°
- **With: flat across angles**
- A little lower upright

A contact map is an image



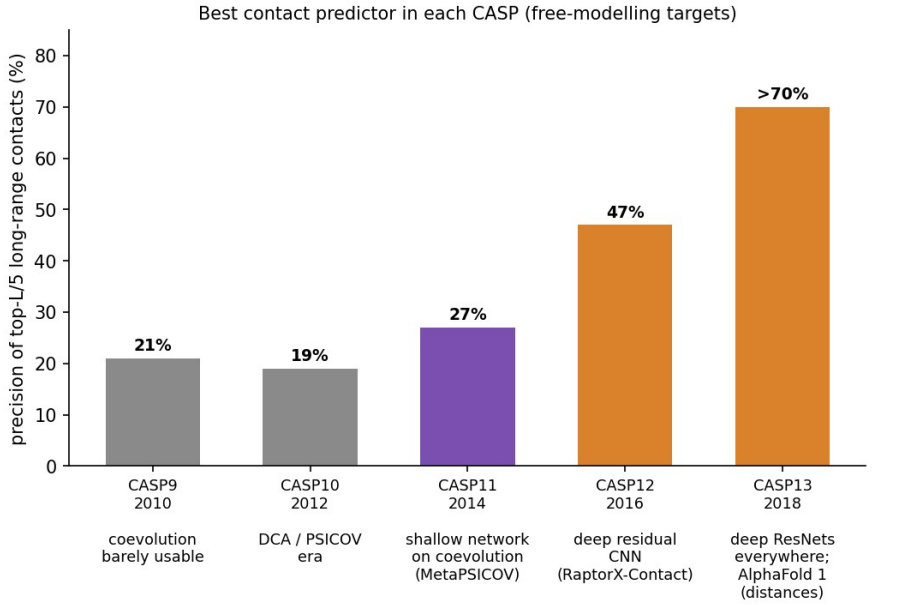


Row i, column j: black if residues i and j are close (Cβ-Cβ < 8 Å). Each contact appears twice, mirrored in the diagonal. Green: within one helix, next to the diagonal. Blue, orange, purple: helices packed against each other, far from the diagonal. Local patterns → 2D convolutions.

PDB 4HHB chain B, rendered with PyMOL; tools/builders/make_session10_contacts.py.

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Deep CNNs on contact maps transformed structure prediction



CASP14 (2020): top-5 average 64%, no further gain; AlphaFold 2 made contact prediction a by-product of full 3D prediction.

Best contact predictor in each CASP. Deep residual CNNs that predict the whole map as an image arrived in 2016 (47%) and passed 70% by 2018. Session 12 tells the rest of the story.

CASP assessments; see Session 12.

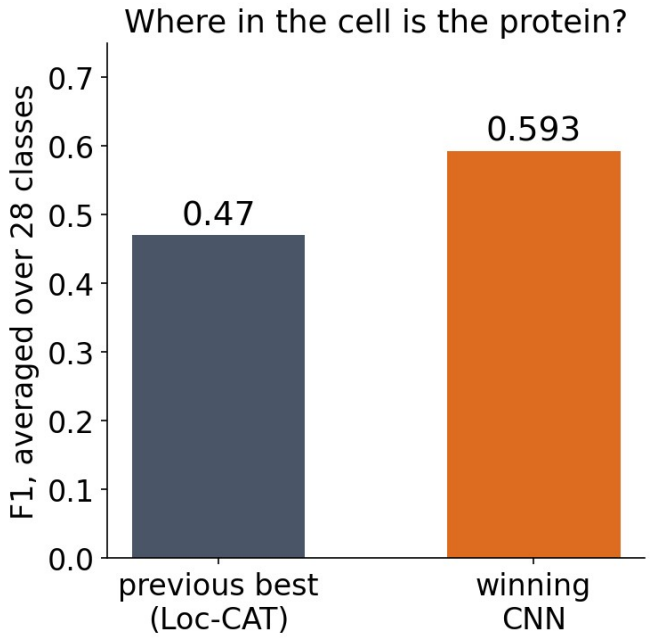
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Where in the cell is the protein? A CNN competition

42,774 fluorescence images of human cells

28 localisation classes, several per image

2,172 teams in 3 months (Kaggle, 2018-19)



- **Human Protein Atlas images**
- Several labels per image, rare classes
- Best CNNs: 26% better
- Still below the experts

Ouyang et al. (2019), Nature Methods 16:1254, 'Analysis of the Human Protein Atlas Image Classification competition'.

Alpha Cell: towards a predictive model of the human cell



starts from the HPA: > 10 million annotated images; single-cell expression in 81 cell types

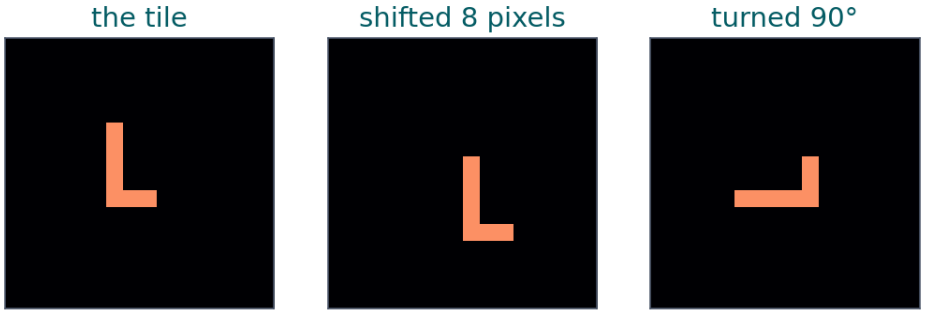
- **SciLifeLab and KTH**
- Led by Jan Ellenberg and Mathias Uhlén
- Knut and Alice Wallenberg Foundation
- **Starts from the Human Protein Atlas**

scilifelab.se/alpha-cell; Human Protein Atlas news, 7 May 2025. https://www.scilifelab.se/alpha-cell/

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Shift the tile, or turn it?

16 random 3 × 3 filters, then global max pooling. Which change alters the 16 pooled features?



16 random 3 × 3 filters, then global max pooling

A Only the shift

B Only the turn

C Both

D Neither: a CNN ignores both

Guess on your own

Scan the code, or go to
quiz.bioinfo.se/live

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Shift the tile, or turn it?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

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Shift the tile, or turn it?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session10-discussion-3.ipynb

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Shift the tile, or turn it?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

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Shift the tile, or turn it?

Results, and who explains

Both rounds side by side.

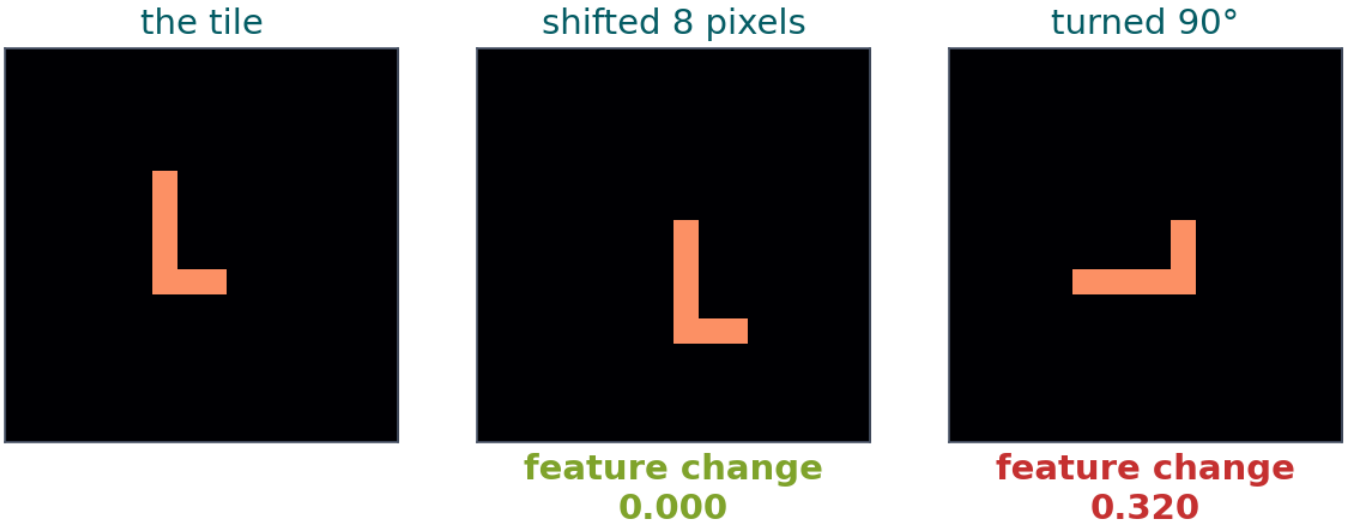
Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

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The answer: only the turn changes the features

Shared filters follow a shift, and pooling forgets where. Nothing undoes a turn.



Shifting moves the object, and the shared filters find the same patterns at the new place; global pooling then forgets where. Nothing in the architecture undoes a turn, so the features change. Rotation invariance has to be learned, for example by augmenting with turned copies, but only where turning keeps the label.

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In up to three words: what was the hardest part of today?

Why convolution	Sequences	Images
weight sharing, equivariance	filters as motifs, pooling	transfer, augmentation, contact maps

Answer now

Scan the code, or go to

quiz.bioinfo.se/live

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The hardest part of today

What the room found hardest

Show the results ▸ quiz.bioinfo.se/live/report

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Session 11 next

Dense: every position has its own weights. CNN: one local detector, reused
But a CNN sees only its receptive field, and global pooling forgets where
Next: models that read a whole sequence in order: RNNs, language models, Transformers